

Tutorial 04: Controller Upgrading

A practical local-server Screeps module

<docs/tutorials/04-controller-upgrading.md>

Episode 4: The Controller's Price

The controller crosses its first threshold while you're not even watching it – like finding out you hit a revenue milestone from a Slack notification instead of a spreadsheet you were staring at. Your collaborator notices before you do.

"Room Controller Level 2," it says. "You've been upgrading it because the upgrader role needed a job to do. Now it's paying you back – extensions just unlocked. More energy capacity per spawn cycle, bigger creeps, everything downstream gets faster. This is what people mean when they say 'compounding.'"

The controller was never just a progress bar. It's the thing that decides what you're allowed to build next, same as revenue decides what headcount you're allowed to add.

Goal

Reach Room Controller Level 2 (RCL2), understand what it unlocks, and spend it:

- Confirm RCL2 and read what it grants
- Place extension construction sites
- Add a `builder` role that completes them
- Watch `room.energyCapacityAvailable` increase as a direct result

Prerequisites

Tutorial 03 is complete:

- `role.harvester.js`, `role.upgrader.js`, and `main.js` exist.
- The `POPULATION` system is spawning 2 harvesters and 1 upgrader.
- The controller's `progress` has been increasing.

Step 1: Check Your Controller Level

```
Game.spawns.Spawn1.room.controller.level
```

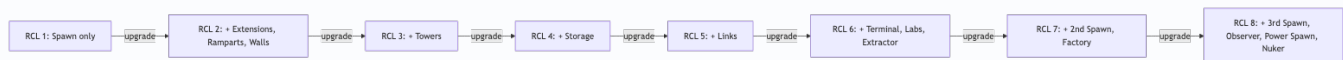
If this is still `1`, let the upgrader keep working – RCL2 requires `progress` to reach `progressTotal` at level 1, which happens automatically as your upgrader delivers energy. This can take a while at one upgrader; that's expected. Checkpoint: this returns `2` before you continue.

Step 2: Understand What RCL Controls

Room Controller Level gates which structures you're allowed to have, not which structures you already have. Reaching a level only unlocks the *allowance* – you still have to build.

	RCL	Extensions Allowed	Notable Unlock
	1	0	Spawn only
	2	5	Extensions, ramparts, walls
	3	10	Towers
	4	20	Storage
	5	30	Links
	6	40	Terminal, labs, extractor
	7	50	Second spawn, factory
	8	60	Third spawn, observer, power spawn, nuker

You just crossed row 2. Five extensions are now legal to build in this room. None exist yet.



Each arrow is "allowed to build," not "built automatically" – crossing RCL2 doesn't place a single extension for you. That's still Step 3.

Checkpoint:

```
Game.spawns.Spawn1.room.energyCapacityAvailable
```

Expected result: 300 – the spawn's own capacity. Extensions haven't been built, so they aren't contributing yet.

Step 3: Place Extension Construction Sites

Pick open tiles near `Spawn1` – not on terrain walls, not on the spawn itself. In the console, using coordinates near your spawn (adjust `x / y` to your room):

```
const room = Game.spawns.Spawn1.room;
room.createConstructionSite(24, 24, STRUCTURE_EXTENSION);
room.createConstructionSite(25, 24, STRUCTURE_EXTENSION);
room.createConstructionSite(24, 25, STRUCTURE_EXTENSION);
room.createConstructionSite(23, 24, STRUCTURE_EXTENSION);
room.createConstructionSite(24, 23, STRUCTURE_EXTENSION);
```

Expected result for each call: 0.

Common results:

- 0: OK
- -7: invalid target (wall, or another structure/site already there) – pick a different tile
- -8: the site limit for this room has been reached

Checkpoint:

```
Game.spawns.Spawn1.room.find(FIND_CONSTRUCTION_SITES).length
```

Expected result: 5 (or however many succeeded).

Step 4: Add the Builder Role

Nobody is completing these sites yet. Add a new file named `role.builder`:

```
const roleBuilder = {
  run(creep) {
    if (creep.memory.building && creep.store[RESOURCE_ENERGY] === 0) {
      creep.memory.building = false;
    }
  }
}
```

```

    if (!creep.memory.building && creep.store.getFreeCapacity() === 0) {
      creep.memory.building = true;
    }

    if (creep.memory.building) {
      const target = creep.pos.findClosestByPath(FIND_CONSTRUCTION_SITES);
      if (!target) {
        return;
      }
      if (creep.build(target) === ERR_NOT_IN_RANGE) {
        creep.moveTo(target, { visualizePathStyle: { stroke: '#ffffff' } });
      }
      return;
    }

    const sources = creep.room.find(FIND_SOURCES);
    if (creep.harvest(sources[0]) === ERR_NOT_IN_RANGE) {
      creep.moveTo(sources[0], { visualizePathStyle: { stroke: '#ffaa00' } });
    }
  },
};

module.exports = roleBuilder;

```

This follows the exact same toggle pattern as `role.upgrader.js` from Episode 3: fill up, switch to "spend" mode, empty out, switch back. Same shape, different target.

Step 5: Wire It Into `main`

Add the builder to the population and the dispatch loop:

```

const roleHarvester = require('role.harvester');
const roleUpgrader = require('role.upgrader');
const roleBuilder = require('role.builder');

const POPULATION = {
  harvester: 2,
  upgrader: 1,
  builder: 1,
};

module.exports.loop = function () {
  for (const name in Memory.creeps) {
    if (!Game.creeps[name]) {
      delete Memory.creeps[name];
    }
  }

  spawnMissingRoles();

  for (const name in Game.creeps) {
    const creep = Game.creeps[name];
    if (creep.memory.role === 'harvester') {
      roleHarvester.run(creep);
    } else if (creep.memory.role === 'upgrader') {
      roleUpgrader.run(creep);
    } else if (creep.memory.role === 'builder') {
      roleBuilder.run(creep);
    }
  }
};

```

```
function spawnMissingRoles() {
  const spawn = Game.spawns.Spawn1;
  if (spawn.spawning) {
    return;
  }

  for (const role in POPULATION) {
    const currentCount = _.filter(Game.creeps, (creep) => creep.memory.role === role).length;
    if (currentCount < POPULATION[role]) {
      const name = `${role}${Game.time}`;
      spawn.spawnCreep([WORK, CARRY, MOVE], name, { memory: { role } });
      break;
    }
  }
}
```

Checkpoint:

- A new `builder` creep spawns.
- It harvests, then walks to the nearest construction site and builds it.
- The site's progress bar (visible in the client) moves.

Step 6: Watch Capacity Change

Let the builder finish at least one extension. Then check:

```
Game.spawns.Spawn1.room.energyCapacityAvailable
```

Expected result: `350 - 300` from the spawn plus `50` from one completed extension. Each additional finished extension adds another `50`, up to `550` once all five are standing.

This number is not cosmetic. `spawnCreep` can only afford body parts up to whatever `energyCapacityAvailable` allows. A colony stuck at `300` capacity is permanently stuck with 3-part creeps like the ones you've been spawning all along.

 **Screenshot placeholder:** Five completed extensions surrounding `Spawn1` in the client, with the room's energy capacity UI visible – the moment "more capacity" stops being a console number and becomes visible structures.

Troubleshooting

If `createConstructionSite` keeps returning `-7`, you're likely targeting a wall tile or a position already occupied. Check terrain first:

```
Game.spawns.Spawn1.room.getTerrain().get(24, 24)
```

`1` means wall. `0` means plain. `2` means swamp (buildable, just slow).

If the builder never switches to building mode, confirm `FIND_CONSTRUCTION_SITES` is actually returning something:

```
Game.spawns.Spawn1.room.find(FIND_CONSTRUCTION_SITES)
```

If `energyCapacityAvailable` isn't increasing after a site completes, confirm the site actually finished and became a real structure:

```
Game.spawns.Spawn1.room.find(FIND_MY_STRUCTURES, { filter: { structureType:
STRUCTURE_EXTENSION } }).length
```

Completion Criteria

You are done when:

- The room is at RCL2 or higher.
- At least one extension construction site has been placed and completed.
- `role.builder.js` exists and is wired into `main`.
- `room.energyCapacityAvailable` is greater than `300`.

Learning Notes

After completing the tutorial, write down:

- How long did it take, in ticks, to go from RCL1 to RCL2 with only one upgrader?
- What's the tradeoff of adding a second upgrader now versus waiting for more extensions first?
- `spawnCreep` still uses the fixed body `[WORK, CARRY, MOVE]` even though capacity went up. What's the smallest change that would let it spend the new capacity?
- The builder's toggle pattern is identical to the upgrader's. What does that similarity suggest about how these two files should relate to each other going forward?

Next: Episode 5 – Paved Roads

The builder has a job now, but it's a job with an end date – five extensions, then nothing left to build.

"That's the wrong way to think about a builder," your collaborator says. "The room isn't going to stop needing things built. It's going to need roads next, so your creeps stop torching half their lifetime walking across open terrain like it's a commute nobody's reimbursing."

See `docs/roadmap.md` for the full season.